

NB SG 4 – Cases of Deep Cervicofacial Spaces

Required materials and equipment:

1. A copy of the clinical cases
2. A copy of the research article
3. An electronic copy or hard copy of your Netter's Atlas (required) and lecture notes (optional)

How to prepare and what to expect from this session

1. Research Article: 15 minutes

Use library resources or open resources (e.g. Google Scholar) to find a peer reviewed article on temporomandibular joint disorders. Use the following objectives as a guide to review the selected paper. Once you are able to answer most of these questions, then you have successfully completed your review and are fully prepared for discussion.

- a) How did you find the article?
- b) Why did you choose the specific article?
- c) What are the typical complaints associated with the syndrome?
- d) With these symptoms what other possible diagnosis should be explored?
- e) What questions did you have while reading the paper?

2. Clinical Case Discussion: 80 minutes

a) Use your lectures, DLAs, textbooks and Sketchfab models to answer the clinical case questions listed below. These questions will be discussed with your peers and clinical facilitators during the small group. At the end of the session, all students are responsible for having completed all the learning objectives listed for that session. If a group is not able to finish all learning objectives in the allotted time, students have the individual responsibility for learning them on their own. If a group finishes all learning objectives before the allotted time has ended, the students must reinforce the learned concepts by inviting interactions with Facilitators, peers and subject experts.

b) How to use SketchFab:

I) Open the links below

a. IT PT Fossae: <https://sketchfab.com/3d-models/nerve-content-of-the-it-ptfossae-6cbe72f8e0fe4e9a9321fc6f81b2e46d>

b. IT PT Fossae Box Model: <https://sketchfab.com/3d-models/content-andboundaries-of-the-it-pt-fossae-2b139ac74a7f4f2a9c218c56029848be>

c. Spaces of the Head and Neck: <https://sketchfab.com/3d-models/spaces-ofthe-head-and-neck-anatomical-modelace11450d00d4653b4c34bf53bfc70a4>

II) Use the mouse or trackpad to rotate and zoom in/out to view the 3D models.

III) Utilize the 3D visualization with the cases in the small group session to better understand the anatomical structures.

Clinical Case Questions:

Case A

Objectives

SOM.MK.I.BPM2.1.NB.1.ANAT.0146	Describe the general attachments, innervations and actions of the muscles of mastication
SOM.MK.I.BPM2.1.NB.1.ANAT.0149	Describe the mandibular nerve, giving its entrance foramen into the fossa, its branches, the nerve fiber content and distribution of each branch
SOM.MK.I.BPM2.1.NB.1.ANAT.0029	Describe how the ophthalmic veins communicate with the pterygoid venous plexus, facial vein, and the cavernous sinus
SOM.MK.I.BPM2.1.NB.1.ANAT.0150	Describe the chorda tympani nerve, its origin, fiber types, connection to the lingual nerve, target ganglion and final distribution.

A 38-year-old woman comes to the physician because of fever, facial pain, restricted mouth opening, swelling in the temporal region and numbness of the right lower lip of 9 days duration. She explains that she had her upper right 3rd molar extracted 3 days before her symptoms started. During the procedure the dentist needed to inject multiple times to achieve the necessary anesthetic effect of the area. A diagnosis of infratemporal fossa abscess is made.

1. What nerve branch injury is the most likely cause of difficulty in opening the mouth?
2. Name the muscles of mastication and their major functions.
3. What nerve injury resulted in the numbness over the lower lip?
4. What other nerve could potentially be damaged due to the abscess? What would be the resulting symptom?
5. Why would this condition be life threatening if left untreated?
6. Explain how the infection can spread from the infratemporal fossa to the cavernous sinus

Case B

Objectives

SOM.MK.I.BPM2.1.NB.1.ANAT.0066	Describe the general attachments, innervation, and actions of the muscles that define the borders of the posterior triangle
SOM.MK.I.BPM2.1.NB.1.ANAT.0071	Describe the cervical plexus, its branches, and their distribution
SOM.MK.I.BPM2.1.NB.1.ANAT.0085	Describe the location, innervation and blood supply of the parotid gland
SOM.MK.I.BPM2.1.NB.1.ANAT.0090	Discuss how pathology of the parotid gland may lead to facial palsy

A 24-year-old man was stabbed in the neck during a brawl, with the knife penetrating the right sternocleidomastoid muscle just posterior to the angle of the mandible. Following hospitalization and treatment, he complained of the following:

- his head was tilted to the left shoulder, with the face turned to the right and upward
- numbness over part of the right auricle and the adjacent scalp
- pain at the wound site
- difficulty with tongue movement

1. What nerve was injured and what muscle caused his face to turn upward to the right?
2. What nerve was severed to result in numbness over the right auricle and scalp?
3. What is the origin of this nerve?
4. What do you believe was injured to result in difficulty in tongue movement?
5. If the knife had penetrated the substance of the parotid gland, which important structures (vessels and nerves) passing through the substance of the parotid gland could have been damaged?
6. Which nerve is responsible for the secretions of the parotid gland?

Case C

Objectives

SOM.MK.II.BPM2.1.NB.5018	Analyze and compare the attachment points, innervations, and functions of the levator palpebrae superioris and superior tarsal muscles
SOM.MK.II.BPM2.1.NB.5020	Compare and contrast the sensory functions carried by the tympanic nerve, chorda tympani, and the vestibulocochlear nerve
SOM.MK.II.BPM2.1.NB.5021	Analyze the tympanic nerve plexus, detailing its fiber types and distributions, while integrating its role in the sensory innervation of the middle ear structures
SOM.MK.II.BPM2.1.NB.5022	Evaluate the anatomical features, innervation, and actions of the tensor tympani and stapedius muscles

A 9-year-old boy is brought to the clinic by his mother because of 5 days of persistent left-sided ear pain and headache. His symptoms began with a mild upper respiratory tract infection 1 week ago, followed by ear discomfort that did not improve despite over-the-counter analgesics. Over the past 2 days, he has increasing pain around his left eye, numbness over his left forehead, and weakness on the left side of his face. His mother reports that he has difficulty closing his left eye and that food tends to spill from the left corner of his mouth while eating. He has no significant past medical history and is up to date with vaccinations. His symptoms began shortly after an episode of acute otitis media treated with a 5-day course of oral antibiotics. His medical history is otherwise unremarkable, and he has no history of trauma.

On examination, his temperature is 38.6°C (101.5°F). Neurologic examination shows reduced sensation over the left forehead and upper cheek, weakness of the muscles on the left side of the face, decreased forehead wrinkling, and incomplete closure of the left eye. weakness in abduction of the left eye and decreased sensation in the ophthalmic division of the left trigeminal nerve. MRI of the head shows inflammatory changes of the petrous portion of the left temporal bone.

1. Which cranial fossa(e) may be affected?
2. Trace the anatomical route by which an infection can spread to the affected cranial fossa(e).
3. Which region of the lateral neck provides a potential pathway for infections originating near the ear to spread into deeper cervical compartments? Describe the anatomical features that facilitate this spread.

4. Explain the sensory pathways responsible for changes in forehead, cheek, and periauricular sensation. Explain the mechanisms underlying referred ear pain from pathology outside the ear.

5. Describe the course of the nerve controlling facial expression within the temporal bone and its branching within the lateral face. Explain how a lesion can produce facial weakness with or without associated taste and secretomotor deficits.

6. Compare the functions and sensory distributions of the chorda tympani and vestibulocochlear nerves.

7. What are the functions and innervations of the tensor tympani and stapedius muscles? How might inflammation in this region contribute to the patient's symptoms?

8. Considering nearby nervous pathways, what specific additional signs or symptoms should be assessed? What additional tests would you perform to help determine disease progression?

9. Compare the motor functions involved in chewing and facial expression. Describe bedside examination techniques that distinguish sensory from motor impairments affecting these functions.